Radial Kernel f_i Optimization Report

27 February 2026

Approach

1. Radial f_i optimization method

The notebook parameterizes a 2D kernel using radial bins:

$$K(x, y) = \sum_{r=0}^{R} f_r \, B_r(x, y), \quad R = 10$$

where B_r are ring-shaped basis masks and f_r are learnable radial coefficients.

Optimization objective:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda_{\text{smooth}} \frac{1}{R} \sum_{r=1}^{R} (f_r - f_{r-1})^2 + \lambda_{\text{anchor}} \frac{1}{R+1} \sum_{r=0}^{R} (f_r - f_r^{\text{init}})^2$$

with $\lambda_{\text{smooth}} = 10^{-3}$ and $\lambda_{\text{anchor}} = 5 \times 10^{-2}$.

Classifier head and training:

- A 1D logistic head (w, b) is trained jointly with f_i using Adam.
- Epochs = 25, batch size = 128, learning rates: 10^{-2} for f_i and 10^{-2} for head parameters.
- Patch standardization is disabled.
- Model selection criterion: maximum eval ACC at threshold 0.5.
- A tuned threshold is also reported, selected on train via grid search.

Results

2. Best checkpoint performance and rotation stability

Using the current notebook outputs:

- Best-checkpoint metrics: **AUC = 0.897333**, **ACC@0.5 = 0.852000**.
- Invariant kernel (rotated every 10°) accuracy: **AUC = 0.875518**, **ACC@0.5 = 0.793143**.

Inspect Learned Coefficients and Kernels

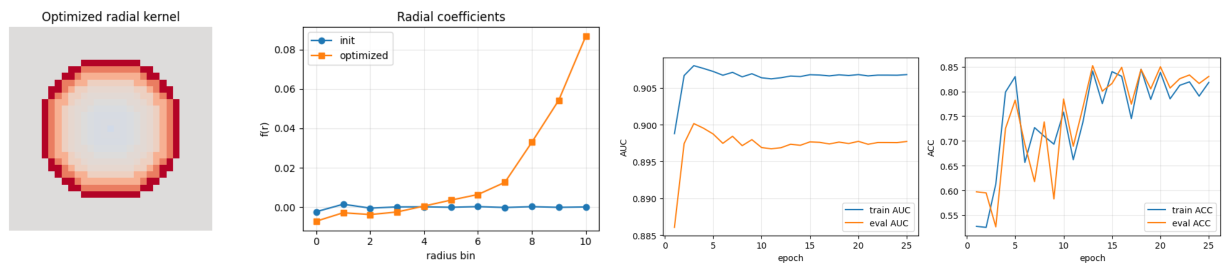

Figure 1: Left: initial radial kernel, optimized radial kernel, and learned $f(r)$ coefficients. Right: training curves for train/eval AUC and ACC across epochs for the trained radial kernel (f_i optimization).

Full-Image Inference + Heatmaps

The figures below correspond to inference with the trained radial kernel (optimized via f_i learning; kernel rotated every 10°).

Set A (Invariant kernel, rotated every 10°):

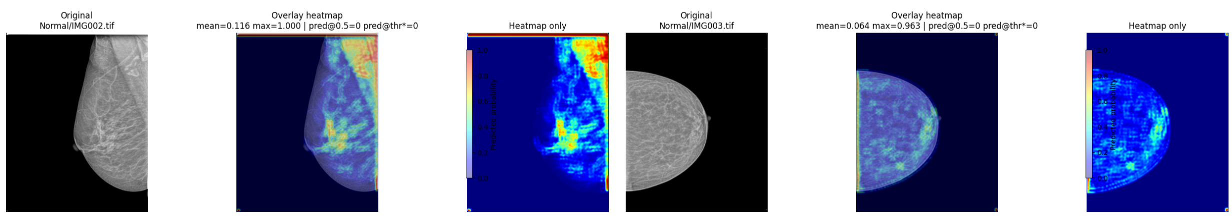

Figure 2: Normal class, Set A examples 1–2 (trained radial kernel, rotated every 10°).

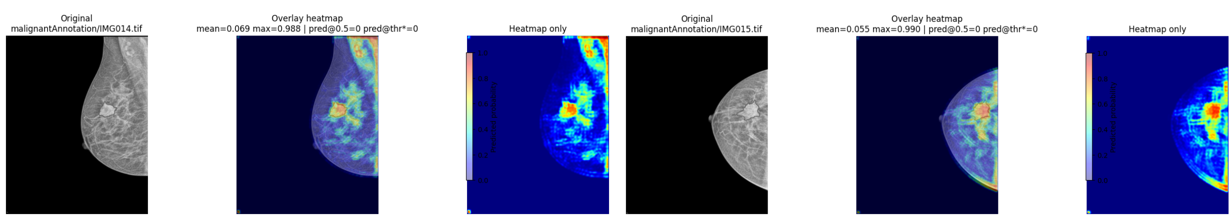

Figure 3: malignant class, Set A examples 1–2 (trained radial kernel, rotated every 10°).

Set B (Trained radial kernel):

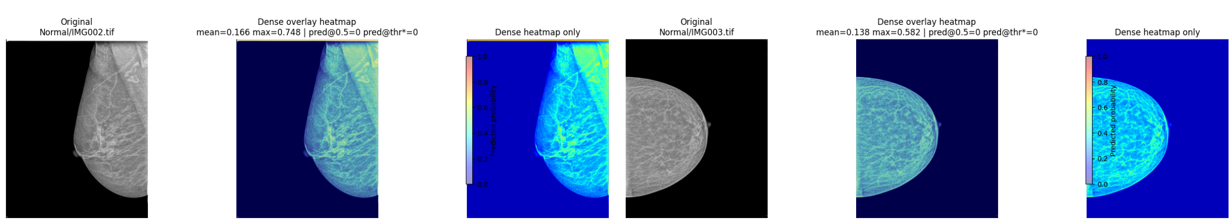

Figure 4: Normal class, Set B examples 1–2 from trained-radial-kernel (f_i) inference.

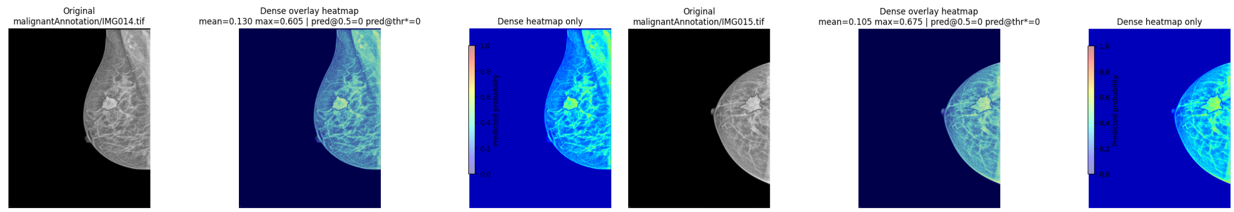

Figure 5: malignant class, Set B examples 1–2 from trained-radial-kernel (f_i) inference.

Findings

- The radial-parameterized optimization of f_i improved performance over the manually created invariant kernel baseline.
- Although the training metrics are very good, the heatmap quality does not perform well qualitatively.

Diagnosis

Several refinements for heatmap behavior:

- Reduced inference stride from coarse overlap to a finer setting.
- Added Gaussian patch-weight blending instead of uniform patch filling.
- Used smoother rendering (bilinear interpolation) and higher save DPI.
- Added a dense full-image inference block (per-pixel response map) to reduce square artifacts from patch-level aggregation.

These changes improved visual smoothness, but the qualitative localization is still weaker than expected in.

Next Steps

- Revisit the response definition for localization (e.g., avoid over-reliance on global max-like behavior).
- Use smaller patch size and/or multi-scale inference to improve spatial detail.
- Add hard-negative mining from normal images that still receive high heatmap scores.
- Evaluate localization with explicit region-level metrics (not only patch-level AUC/ACC).

Conclusion

The trained radial-kernel approach with f_i optimization gives strong quantitative performance in this run (best Eval AUC = 0.897333, ACC@0.5 = 0.852000), and it outperforms the invariant-kernel rotated-every-10° baseline (AUC = 0.875518, ACC@0.5 = 0.793143) while remaining stable across right-angle rotations. However, despite these good training/evaluation metrics, the qualitative full-image heatmaps are still not satisfactory. In summary, the method is promising for classification accuracy, but heatmap quality/localization needs further improvement before it can be considered reliable for visual interpretation.